Fulton Hogan Assignment

Software Modelling Report ENSE706

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1. Problem Statement

Fulton Hogan is a New Zealand based construction company that provides quality infrastructure. They are known for constructing and maintaining roads, bridges, energy and water networks all across New Zealand and Australia. Their goals outside of construction have always been “creating, connecting, and caring for communities”.

As of now, Fulton Hogan’s Project Costing & Management section uses JD Edwards to manage their main operations. JD Edwards is an Enterprise Resource Planning system which manages construction projects, equipment, and payroll (Oracle, 2024). The users of this system use it to perform crucial business operations that are important for the successful running of the business. However, it is an old legacy system that is not meant to handle real-time demands of today’s construction management needs.

Each department uses JD Edwards in different ways, however it cannot keep up with modern business-related demands. Fulton Hogan is looking to improve their Project Costing & Management System, which involves three departments; Project Management department, Finance department, and Operations department.

1.1 Project Management Department

The main responsibilities of this department are:

- Creating and managing projects
- Logs timesheets, milestones, and project progress
- Submits budget requests and cost updates

The Project Management department serves as the primary driver of project planning and execution. They are responsible for the entire process of a construction project from start to finish. This department ensures projects are on schedule, and that any tasks are completed within the time given. However, all users of this system have reported latency issues. Whenever Project Coordinators contribute to the project, Project Managers cannot see real-time progress meaning the Site Leads don’t know what to do in current time. This is due to the outdated legacy program, causing scheduling conflicts and data inconsistences.

The main users of this department are Project Managers, Site Leads and Project Coordinators. Project Managers are the people that manage the entire project. When given a project, they are responsible for managing both people and resources. Site Leads are the people who are on the construction sites and oversee everything going on, making sure everything is following the correct procedure. Project Coordinators are the people working on the project and

contributing their input. They are always finding solutions to any problems that arise, making sure everything is working smoothly, and acting as a central point of communication.

The Project Management Department uses the system as the central hub for data entry and project tracking. They initialise projects, log any progress made, and any updates are also triggered by these three user roles. Each user has different privileges which allow them to access different parts of the system. Project Managers can initialise projects and add any major changes to them (extending deadlines, generating team reports, managing the team). However, they cannot see what the Site Leads are telling the Site Foremen. Site Leads use the system to send instructions to the Site Foremen on duty. They need all information available to them and the tasks that are needed to achieve in a selected time period. Project Coordinators use the system to document any new findings, managing new information, analysing data, and creating reports for the Project Manager.

1.2 Finance Department

The main responsibilities of this department are:

- Approving and monitoring project budgets
- Tracking actual vs estimated costs
- Processes financial reporting on projects

The Finance department is responsible for overseeing the financial aspects of project delivery. They ensure that the company is always making profits by reviewing every dollar allocated to the project. This department is always focused on financial data and making sure everything on the financial side is going smoothly. Auditors are always complaining about missing data every now and then. The data was entered, but the auditors could not see the data. This means the data loss was a system error. Fulton Hogan wants this problem to be sorted as a high priority. This is because the data is read by people who make important decisions that lead the project progress forwards.

Financial Controllers, Project Accountants, and Auditors are users for this department. Financial controllers manage the day-to-day financial operations such as monthly reporting, managing cash flow, and ensuring accounting standards are met across project timelines. Project Accountants have the main responsibility of handling financial tracking for projects. Auditors are always reviewing financial records and processes to verify financial accuracy when it comes to project resources.

The Finance Department uses the system as the primary platform for financial analysis and reporting, making sure all expenses remain aligned with approved budgets. They are always comparing Estimated vs Actual (EVA) costs, approving the funds for Project Management requests, and generating monthly compliance reports. Similar to the Project Management Department, the users in this department have different privileges on the system. The

Financial Controllers have access to all resources so that they can make profitable decisions. Project Accountants have access to tracking software, however not to the reports that show project spendings. Auditors have access to all financial reports, because they need to have all financial information to create reports for the Project Managers.

1.3 Operations Department

The main responsibilities of this department are:

- Executing the on-ground construction work
- Reporting labour hours and equipment usage
- Flagging project issues that affect cost or timeline

The Operations department is responsible for handling all the physical execution, heavy machinery, labour and materials. Their responsibility is to safely and efficiently deliver the infrastructure for their clients. All users of JD Edwards in this department consistently report issues with system performance. Site Foremen are always having to wait long periods of time for instructions to come through. This slows down progress on the site, since no one can go forward without instructions. Workers have experienced incorrect incomes because their hours weren't logged on properly. Fulton Hogan wants to find the system part responsible for this and remake it.

The main users of this department are the Site Foremen, Heavy Equipment Operators, and General Laborers. The Site Foremen are the supervisors of this department, always managing the day-to-day activities of all the workers. They are constantly translating instructions given by the Site Leads, and other higher ups. Heavy Equipment Operators are in charge of operating all the tools, vehicles, and equipment used to deliver the infrastructure. They use the system for communicating with the Site Foremen and General Laborers, and report anything to do with the equipment and machines. However they should also have access to the instructions given by the Site Lead. This will ensure they are prepared for the upcoming tasks that day and can plan ahead. The General Laborers are the entry level workers who carry out physical tasks on site. They use the system for communication, reporting any problems, and logging in their hours.

This department uses the system as a daily activity log, where everything that happened in the work day is reported and given to people back at the company. The data reported by this department include how many people were working, state of the machines, work to be done the next day, etc. They also have a channel dedicated to reporting issues e.g. broken vehicle or machine.

2. System Overview

Fulton Hogan wants to develop a completely new and updated Project Costing & Management System that allows all three departments to function effortlessly and efficiently. The system should allow any data inputted to go through successfully and without issue, at the same time the other devices using this software should be updated instantaneously. All users should have the correct amount of access, while some users get more privileges than others depending on their role in the company. Others should have updated privileges from the old system, and have more access to important documents that will help them work more effectively. The new Project Costing & Management system should improve efficiency, reduce the time it takes for others to be notified of changes to any data or projects, and provide a much better experience to all system users. This will be done by applying modern software principles that can meet the demands of modern project management.

Below is the table describing all stakeholders of the system.

Stakeholder	Roles	Department
Project Manager	Initialise projects, Manage projects, Generate team reports, Manage the project management team, Receive financial reports from Finance department	Management
Project Coordinator	Log new findings, Manage new information, Analyse data, Create progress reports	Management
Site Lead	Guide Site Foremen during work, Have access to day-to-day instructions, Communicate with project management team	Management
Financial Controller	Access to all financial resources, Make final decisions	Finance
Project Accountant	Access to financial tracking software, Prepare financial statements	Finance
Auditor	Access to all financial reports, issue any financial problems, verify financial accuracy, generate compliance reports	Finance
Site Foremen	Manage Instructions from Site Lead to general labourers, flag any issues	Operations

Heavy Machine Operator	Operate machines, communicate with Site Foremen, Log in hours into timesheet, Access to instructions from Site Lead	Operations
General Labourer	Physical activities, Log in hours into timesheet, flag any issues to the department	Operations

2.1 System Scope

The new Project Costing & Management System should improve real-time data entry, tracking, and reporting across all 3 departments at Fulton Hogan: Project Management, Finance, Operations. Everything mentioned about the departments above will be key features in the scope: Project initialisation and management, timesheet logging, budget management, cost tracking, financial reporting, equipment reporting, issue flagging for equipment, and role-based privileges for the different users of the system.

2.2 System Boundaries

This system is limited to the internal operations of Fulton Hogan's Project Costing & Management function. It will not replace or directly integrate with any other external systems that are used by Fulton Hogan or their users. Human Resources, recruitment, and employee reviews are examples of functions that fall out of the scope of this system. The only functions that are being improved are the Project Management department, Finance department, and Operations department. All other business functions fall outside the scope of this system, and are managed by other software. This was confirmed with the client through consultation during the first meeting. Based on the consultation, it has been determined that all requirements have been captured within the scope. There are no other missing requirements.

2.3 Assumptions

The following assumptions have been made during the analysis of this system.

1. All users of this new system will have access to a computer (personal or company provided), along with a stable internet connection. This is needed because the system has real-time operations that require a strong connection to see the new updated information.

2. Fulton Hogan will provide detailed training to all of their employees (managers, labourers, office workers) to switch over to the new system, once it has passed testing and has been deployed.
3. Each user will have an account that is registered to them from JD Edwards. All of their data will go through to the new system and be securely stored, so that new accounts do not have to be made. On top of this, there will be a requirement for the user to have a Fulton Hogan specific email, as any other email will be blocked from logging into the new system.
4. Each user role will have their corresponding privileges within the system, these privileges are described in the Problem Statement.
    a. Users in the Project Management department will have exclusive access to tools linked with project management.
    b. Users in the Finance department will have access to various financial tools, allowing them to analyse financial data.
    c. Users in the Operations department will have a timesheet subsystem, along with a flagging issues page, and more.
5. Existing data stored in JD Edwards will be migrated to the new system without any significant losses. It should be noted that Fulton Hogan is committed to maintaining data integrity during the transition period.

3. Requirements Engineering

This section defines both functional and non-functional requirements of the new Project Costing & Management System for Fulton Hogan. Functional requirements describe what the system must do, these are important for the functionality of the system. Non-functional requirements describe the quality standards the system must meet. This could be performance related like update speeds, or it could be related to security and reliability.

3.1 Requirements Elicitation (Interviews and Observations)

Stakeholder: Project Manager

Q: **What issues do you face with the current system?**
A: The current system was good before, but nowadays, it is very slow. We have to usually wait a minute or two before the new info comes in from our team, to analyse it.

Q: **How do these issues affect your work?**
A: It stops us from moving forward you know? Having to wait for our subordinates' work is a pain, because we can’t do anything without it. When the system does this, we all get annoyed.

Q: **If you were to get a new system to use, what changes would you want on it?**
A: Definitely good performance, the biggest issue with JD Edwards is the speed. It takes a while to see updated information on projects. The new system should show these changes very fast, maybe 10 seconds max.

Stakeholder: Project Coordinator

Q: **What issues do you face with the current system?**
A: So there are actually quite a few issues right now. When I update my work and research onto the system, it takes a while to upload which is quite a pain. I reckon it shouldn't take this long, 10 seconds max would be nice.

Q: **How often do you upload your work onto the system?**
A: We are always uploading stuff whenever we are working, so it could be 2 to several times a day. Which is why it would be nice to have faster uploads.

Q: **If you were to get a new and updated system for you to use at work, what changes would you have?**
A: Definitely a solution to the upload speeds. While this isn't a problem, I reckon it would be nice to have an updated UI. Our one is very old, and feels like we went back two decades in technology.

Stakeholder: Project Accountant

Q: **What issues do you face with the current system?**
A: The system is very unreliable when it comes to tracking actual versus estimated costs. Me and the rest of my team enter financial data throughout the day, but the Auditors come back to us saying they cannot see some of the entries. From our end, everything was done correctly, but the system just causes problems.

Q: **How does this affect your day-to-day work?**
A: It's really frustrating. Imagine you are just doing your work, and you have to submit something to your manager. Then they say "Where is the work you were meant to submit by 5pm?". Yeah, it just slows everything down and causes pain to everyone.

Q: **If you were to get a new system, what would you want improved?**
A: We need our entries to be saved and visible to anyone who needs it. When I log something in, the Auditors should be able to see it straight away. Also an updated GUI would make a nice change in environment for me.

Stakeholder: Auditor

Q: **What issues do you face with the current system?**
A: A big one is missing data. I'm always dealing with missing financial entries in the monthly financial reports for a project. Our accountants did confirm they entered the data, so it's definitely a system issue not a human one.

Q: **How urgent is this issue for your team?**
A: This is a very high priority. The decisions made from the information we review have a huge impact on our projects. Project Managers and Financial Controllers rely on my reports and when they are not complete, progress is slowed.

Q: **What would you want from the new system?**
A: Every financial transaction must be captured without fail. A proper history of who added what entry would make it easier to track where the entry came from. Also we would want to have access to all financial reports so that we can anaylse all information and make our reports more precise.

Stakeholder: Site Foreman

Q: **What issues do you face with the current system?**
A: Well, most of our problems are usually the waiting times for receiving our instructions. Without the instructions from the Site Lead, we cannot move forward.

Q: **How does this affect the work on site?**
A: Productivity goes down a ton. Having to wait when you are all ready to start a task demotivates all the people out here. Everything here depends on those instructions coming to us when needed.

Q: **What would you want from the new system?**
A: Communication needs to be instantaneous. Receiving instructions when we ask should be consistent. A notification for when new instructions arrive would help us out alot aswell. The system needs to be reliable every day, not just sometimes.

Stakeholder: General Labourers

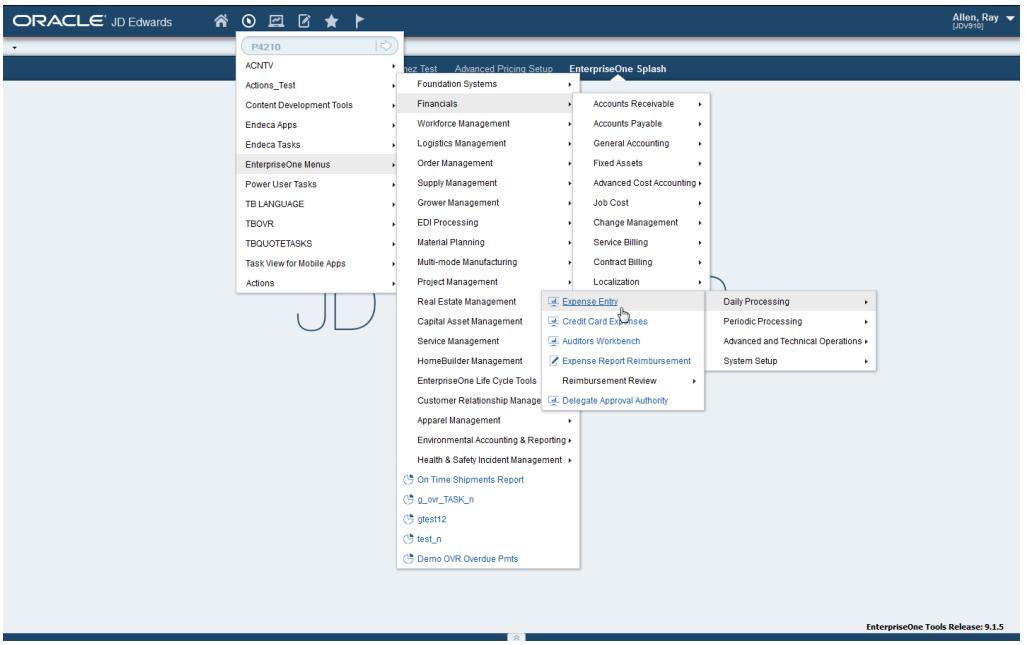
Q: **What issues do you face with the current system?**
A: My hours have not been logging on properly on the system. I enter them at the end of my shifts like I'm told to, but then the pay I get doesn't match them.

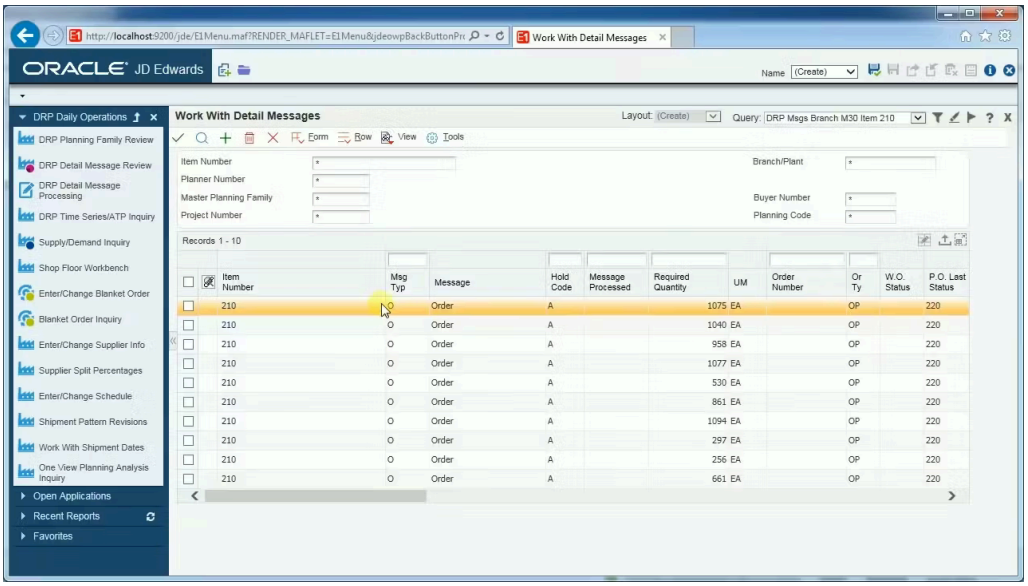
Q: **How do you and your colleagues deal with this?**
A: We usually have to contact the project management team, who then let the financial department know. They usually solve this in around 2 days, but even they are getting sick of it.

Q: **What would you want from a new system?**
A: All I want is for my hours to be logged in the system properly and correctly. I'm out here working in the hot sun, so when I get home, all the money I worked for should be there.

Observations

We could not observe the actual system, however there were online resources that displayed what JD Edwards looks and functions like.





3.2 Functional Requirements

Project Management Department:

Description	Stakeholder
The system shall allow Project Managers to initialise a new project.	Project Manager
The system shall allow Project Managers to modify project details.	Project Manager
The system shall allow Project Managers to manage team members.	Project Manager
The system shall allow Project Managers to view real-time progress updates from all departments	Project Manager
The system shall allow Project Coordinators to log progress, findings, and data entries.	Project Coordinator
The system shall allow Project Coordinators to generate reports of their work.	Project Coordinator
The system shall allow Site Leads to view all project tasks and instructions in real time.	Site Lead
The system shall allow Site Leads to communicate instructions directly to Site Foremen.	Site Lead

Finance Department:

Description	Stakeholder
The system shall allow Financial Controllers to approve all project budgets.	Financial Controller
The system shall allow Financial Controllers to access all financial resources.	Financial Controller
The system shall allow Project Accountants to access financial tracking tools.	Project Accountants
The system shall allow Auditors to access all financial reports for verification.	Auditors
The system shall generate monthly financial reports for the Finance Department	Project Accountants and Financial Controllers

The system shall track and compare estimated versus actual project costs in real time.	Project Accountant

Operations Department:

Description	Stakeholder
The system shall allow Site Foremen to receive and view instructions from Site Leads in real time.	Site Foremen
The system shall allow Site Foremen to relay instructions to General Labourers	Site Foremen
The system shall allow Heavy Equipment Operators to log equipment usage and condition	Heavy Equipment Operators
The system shall allow Heavy Equipment Operators to view instructions from Site Leads	Heavy Equipment Operators
The system shall allow General Labourers to log daily working hours.	General Labourers
The system shall allow General Labourers to report on-site issues through the system.	General Labourers
The system shall provide a dedicated issue flagging channel for reporting equipment faults.	Operations Department

3.3 Non-Functional Requirements

Performance:

Description	Stakeholder
The system shall synchronise data across all departments in under 10 seconds.	All Departments
The system shall support a minimum of 2000 concurrent users without the performance degrading.	All Users
The system shall process and display timesheet entries within 10 seconds of submission. This is what the developer thinks is a reasonable time, to ensure correct logging.	Operations Department

Reliability:

Description	Stakeholder
The system shall maintain a minimum uptime of 99.9% during business hours.	All Departments
The system shall prevent data loss during processing and transmission between departments.	Finance Department and Operations Department
The system shall maintain a full audit log of all data entries and modifications.	Finance Department
The system shall accurately process and record all submitted labour hours without data loss.	Operations Department

Security:

Description	Stakeholder
The system shall encrypt all data transmitted between users and server to hide information from outsiders.	All Users
The system shall restrict access to system features based on the user's role in the company.	All Users
The system shall only allow users with a verified Fulton Hogan email address to sign into the system.	All Users
The system shall comply with the New Zealand Privacy Act 2020.	All Users

Usability:

Description	Stakeholder
The system shall provide tailored user interfaces for each department displaying the tools required for each of them.	All Departments
The system shall show an error message that is in simple English, when an unpredicted input has been entered.	All Users

The system shall provide a modern interface that reduces training time compared to the legacy JD Edwards system.	All Departments
The system shall allow General Labourers to view a summary of their logged hours for the current pay period.	General Labourers

Maintainability:

Description	Stakeholder
The system shall be designed using modern software principles in a modular structure, so a bugged part of the system doesn't affect other parts.	All Departments
The system shall support data migration from JD Edwards without having to manually enter information again.	All Departments

3.4 Requirements Traceability

Req ID	Requirement Description	Stakeholder	Source
FR-PM-01	The system shall allow Project Managers to initialise a new project.	Project Manager	Problem Statement 1.1
FR-PM-02	The system shall allow Project Managers to modify project details.	Project Manager	Problem Statement 1.1 and Project Manager Interview
FR-PM-03	The system shall allow Project Managers to manage team members.	Project Manager	Problem Statement 1.1
FR-PM-04	The system shall allow Project Managers to view real-time progress updates from all departments	Project Manager	Problem Statement 1.1 and Project Manager Interview
FR-PM-05	The system shall allow Project Coordinators to log progress, findings, and data	Project Coordinator	Problem Statement 1.1 and Project Coordinator

	entries.		Interview
FR-PM-06	The system shall allow Project Coordinators to generate reports of their work.	Project Coordinator	Problem Statement 1.1 and Project Coordinator Interview
FR-PM-07	The system shall allow Site Leads to view all project tasks and instructions in real time.	Site Lead	Problem Statement 1.1
FR-PM-08	The system shall allow Site Leads to communicate instructions directly to Site Foremen.	Site Lead	Problem Statement 1.1
FR-F-01	The system shall allow Financial Controllers to approve all project budgets.	Financial Controller	Problem Statement 1.2
FR-F-02	The system shall allow Financial Controllers to access all financial resources.	Financial Controller	Problem Statement 1.2
FR-F-03	The system shall allow Project Accountants to access financial tracking tools.	Project Accountants	Problem Statement 1.2 and Project Accountant Interview
FR-F-04	The system shall allow Auditors to access all financial reports for verification.	Auditors	Problem Statement 1.2 and Auditors Interview
FR-F-05	The system shall generate monthly financial reports for the Finance Department	Auditors and Financial Controllers	Problem Statement 1.2 and Auditors Interview
FR-F-06	The system shall track and compare estimated versus actual project costs in real time.	Project Accountant	Problem Statement 1.2 and Project Accountant Interview
FR-O-01	The system shall allow Site Foremen to receive and view instructions from Site Leads in real time.	Site Foremen	Problem Statement 1.3 and Site Foreman Interview
FR-O-02	The system shall allow Site Foremen to relay instructions	Site Foremen	Problem Statement 1.3 and Site

	to General Labourers		Foreman interview
FR-O-03	The system shall allow Heavy Equipment Operators to log equipment usage and condition	Heavy Equipment Operators	Problem Statement 1.3
FR-O-04	The system shall allow Heavy Equipment Operators to view instructions from Site Leads	Heavy Equipment Operators	Problem Statement 1.3
FR-O-05	The system shall allow General Labourers to log daily working hours.	General Labourers	Problem Statement 1.3 and General Labourer Interview
FR-O-06	The system shall allow General Labourers to report on-site issues through the system.	General Labourers	Problem Statement 1.3
FR-O-07	The system shall provide a dedicated issue flagging channel for reporting equipment faults.	Operations Department	Problem Statement 1.3
NFR-P-01	The system shall synchronise data across all departments in under 10 seconds.	All Departments	Project Manager Interview and Project Coordinator Interview
NFR-P-02	The system shall support a minimum of 2000 concurrent users without the performance degrading.	All Users	Developer's decision of expected organisational user base
NFR-P-03	The system shall process and display timesheet entries within 10 seconds of submission. This is what the developer thinks is a reasonable time, to ensure correct logging.	Operations Department	Developer's decision, following the project manager's 10 second requirement.
NFR-R-01	The system shall maintain a minimum uptime of 99.9% during business hours.	All Departments	Problem Statement 1.1, 1.2, 1.3 all report issues.
NFR-R-02	The system shall prevent data loss during processing and transmission between	Finance Department and Operations Department	Problem Statement 1.2, Auditor Interview and

	departments.		General Labourer Interview
NFR-R-03	The system shall maintain a full audit log of all data entries and modifications.	Finance Department	Problem Statement 1.2 and Auditor Interview
NFR-R-04	The system shall accurately process and record all submitted labour hours without data loss.	Operations Department	Problem Statement 1.3 and General Labourer Interview
NFR-S-01	The system shall encrypt all data transmitted between users and server to hide information from outsiders.	All Users	Problem Statement 1.2. Data loss is a high priority
NFR-S-02	The system shall restrict access to system features based on the user's role in the company.	All Users	Problem Statement 1.1, 1.2 and 1.3. All departments have different uses for the system
NFR-S-03	The system shall only allow users with a verified Fulton Hogan email address to sign into the system.	All Users	System Assumption 3
NFR-S-04	The system shall comply with the New Zealand Privacy Act 2020.	All Users	Employment New Zealand (2025)
NFR-U-01	The system shall provide tailored user interfaces for each department displaying the tools required for each of them.	All Departments	Problem Statement 1.1, 1.2, 1.3. Each department uses the system for different tasks.
NFR-U-02	The system shall show an error message that is in simple English, when an unpredicted input has been entered.	All Users	Developer's decision. Good practice for system use
NFR-U-03	The system shall provide a modern interface that is easier to use, and reduces training time compared to the legacy JD Edwards system.	All Departments	Project Accountant Interview and Project Coordinator Interview
NFR-U-04	The system shall allow	General Labourers	General Labourer

	General Labourers to view a summary of their logged hours for the current pay period.		Interview and Problem Statement 1.3
NFR-M-01	The system shall be designed using modern software principles in a modular structure, so a bugged part of the system doesn't affect other parts.	All Departments	Problem Statement. Legacy system failures affect all departments.
NFR-M-02	The system shall support data migration from JD Edwards without having to manually enter information again.	All Departments	System Assumption 8

4. UML Requirements Modelling

In this section, we will be covering the entire system scope using the Use Case diagram. This type of diagram shows how users interact with a system in a way that the stakeholder can understand comprehensively. There are four main components in the Use Case diagram.

- **Actors:** The users or external entities that interact with or support the system. E.g. Accountant, Manager, Customer.
- **Use-cases:** The actions the system can perform. E.g. "Generate report", "Send Instructions", "Update document".
- **System Boundary:** This is a box that showcases what is inside the system, and what is external.
- **Relationships:** The lines and arrows connecting actors to use-cases. It shows which actor can perform which use-case.

One use-case diagram represents the entire Project Costing and Management group of Fulton Hogan. It shows all the relationships between the actors and use-cases.
The actors in the use-case diagram are: Project Manager, Project Coordinator, Site Lead, Financial Controller, Project Accountant, Auditor, Site Foreman, Heavy Machine Operator, General Labourer.

4.1 Use-Case Descriptions

ID	**UC-01**
Name	Initialise Project
Description	The Project Manager can create a new project using the system Project Management tools. They can set the details and assign team members.
Pre-condition	Users must be logged onto the system, and be a Project Manager currently working for Fulton Hogan.
Event flow	1. Project Manager uses the create button to go onto the "create project" page. 2. Set up the project details, e.g. project title, members, goals, tasks, etc. 3. Submit the project to the projects database
Extension points	If "finish" is clicked while there are missing fields, there should be an error telling the Project Manager to fill them out before creating the project.
Triggers	When a Project Manager is logged into the new system, and clicks the "Create Project" button.
Post-condition	A project is created, and team members can access and contribute to it.

ID	**UC-02**
Name	Verify Financial Accuracy
Description	An auditor views financial reports and goes through them to make sure every detail is accurate and correct.
Pre-condition	The user must currently be an Auditor employeed by Fulton Hogan in the Finance department. Then they access the financial reports database.
Event flow	1. Auditor starts a compliance report 2. Selects a financial report generated by a Project Accountant 3. Goes through the report and verifies all information is correct and accurate. 4. Can flag financial issue if there is any incorrect information
Extension points	If there is a financial inconsistency, the Auditor will "flag" the issue to the system, so that the Project Accountants can be aware of it.

Triggers	When an Auditor starts working on a compliance report.
Post-condition	The compliance report is complete and submitted to the financial reports database.

ID	**UC-03**
Name	Access Financial Resources
Description	The Financial Controller uses all financial tools and information by receiving access by the system.
Pre-condition	Users must be logged onto the system, and be a Financial Controller currently working for Fulton Hogan.
Event flow	1. Financial Controller must be logged into the system 2. Go through reports and tools in the finance departments system page
Extension points	When accessing financial resources, the Financial Controller may want to access the Financial Tracking tool.
Triggers	When a Financial Controller is logged into the system, and has clicked on “All Financial Resources”.
Post-condition	The Financial Controller can choose to approve or dissapprove a budget based on the information he got from the resources.

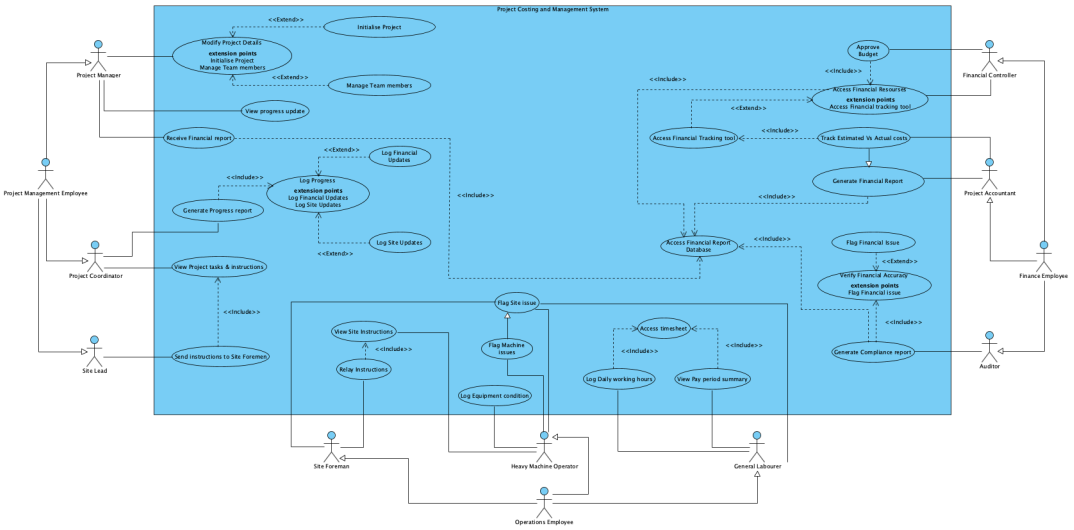
ID	**UC-04**
Name	Log Progress/Findings
Description	A Project Coordinator contributes to the project by uploading new information important to the project, or updating current information.
Pre-condition	The Project Coordinator must be logged into the system. Must first start a “progress report”.
Event flow	1. Project Coordinator must be logged into the system 2. Start a new report 3. Do some research and find updates on the project progress 4. Upload onto the system
Extension points	When contributing new information about the project to the system, they can choose to log in financial information, or onsite

	information.
Triggers	When a Project Coordinator has created a new report.
Post-condition	New information is in the report and uploaded onto the system.

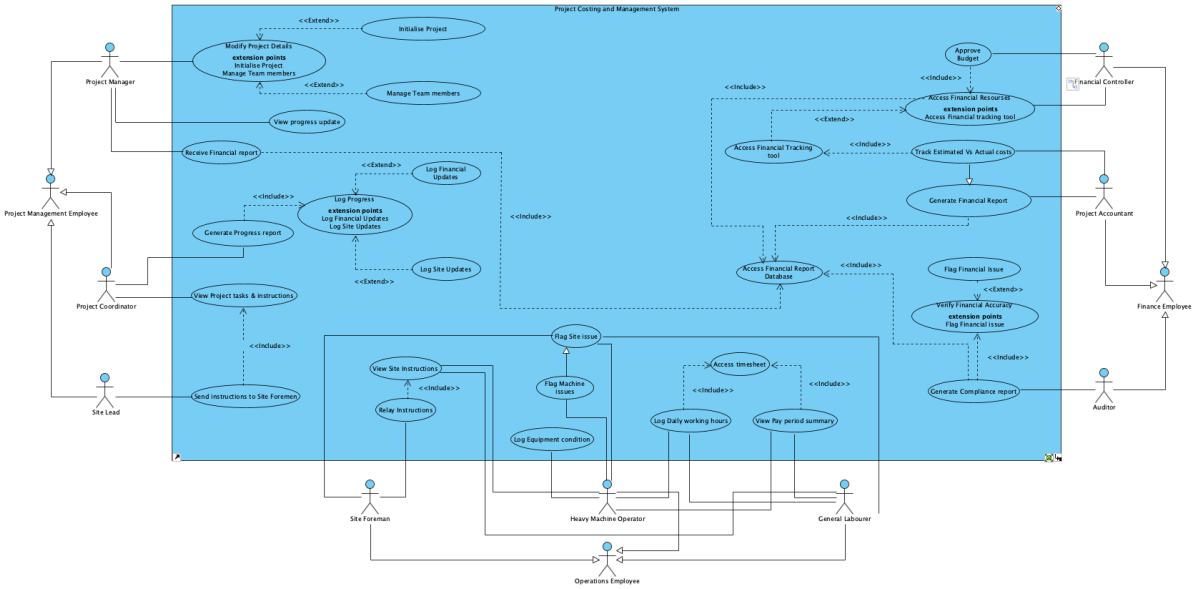
ID	**UC-05**
Name	Log Daily Working hours
Description	A General Labourer enters their hours worked during the day onto the system.
Pre-condition	General Labourer must be logged into the system, and have accessed the timesheet.
Event flow	1. Click “Log Hours” from the Site departments page 2. Enter the amount of hours worked 3. Labourer submits the entry 4. System checks and confirms entry
Extension points	If the pay does not match the hours worked, the General Labourer can go to the Finance department. There they can submit an issue regarding their pay.
Triggers	When a General Labourer clicks “Log Hours” in the system.
Post-condition	Entry is updated in the system, and can be viewed in the Pay Period summary.

4.2 Use-Case Diagram

Iteration One:



Iteration Two:



These use-case diagrams represent the entire Project Costing & Management system for Fulton Hogan. It contains three generalisation relationships. Management Employee generalises Project Manager, Project Coordinator, and Site Lead. Operations Employee generalises Site Foreman, Heavy Machine Operator, and General Labourer. Finance Employee generalises Financial Controller, Project Accountant, and Auditor. This shows the shared behaviour across roles within the different departments while keeping the role-based use-cases separate.

Extend relationships <<extend>> are used where optional behaviour happens under a specific condition. For example, Log Financial Updates and Log Site Updates both extend Log

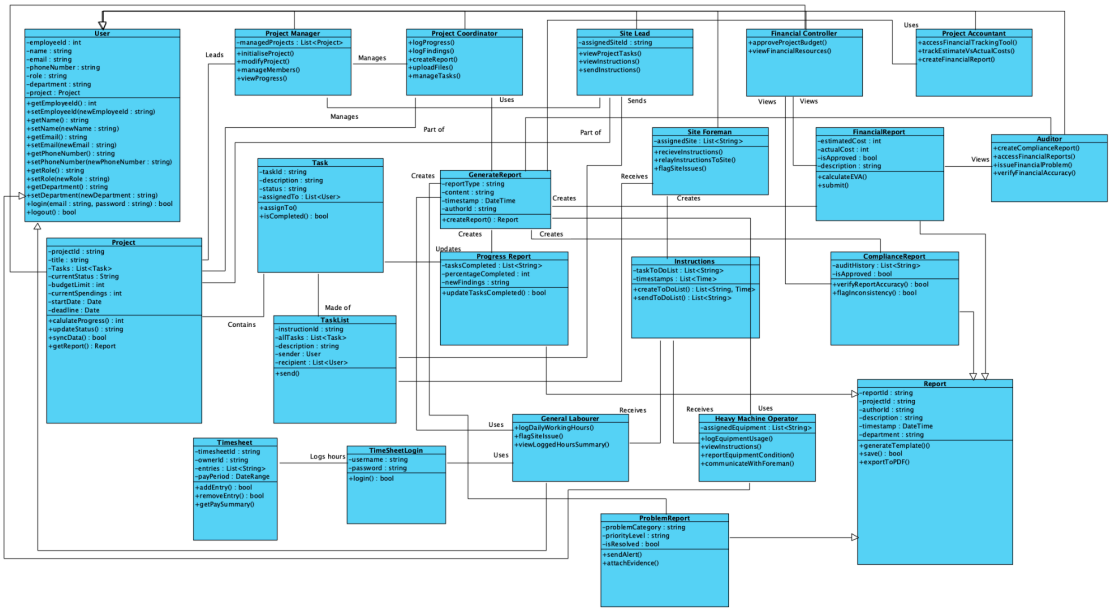
Progress, because the two can happen on an occasion. Therefore it is a conditional action that occurs in specific circumstances.

Include relationships <<include>> are used when a certain use-case involves another use-case all the time. For example, Log Progress includes Generate Progress Report, meaning that to Log Progress, you also need to create a progress report and submit it. Access Financial Resources includes Access Financial Report Database, this is because to access the different resources, you need to retrieve data from the database.

5. Initial Design & Implementation

Using all the work that has been done on planning the Project Costing & Management System, the initial design has started with the first iteration of the Class Diagram.

5.1 Class Diagram - Iteration One



This is the first iteration of the Class Diagram for Fulton Hogan’s Project Costing & Management System. It showcases all the planned C# classes alongside their methods and attributes, relationships between classes, and contains clean mapping.

5.2 Design Weaknesses, Limitations and Risks - Iteration One

Weaknesses:

Lots of classes in the Class diagram are missing reference attributes that are necessary to support their relationships. An example of this is the GenerateReport class, it references a report but lacks the ProjectId attribute. This means that there is no traceable link between a report and the project it belongs to. In Fulton Hogan's system where reports are connected to different projects, this mistake would cause data integrity issues, and make it unreliable to retrieve reports for a certain project.

Currently, the User class is acting as a monolithic class. With no interfaces separating the user roles, this breaks the Interface Segregation Principle. Each of the roles have different responsibilities, and are from different departments. Adding interfaces would better separate the users and will make sure there is security and boundaries between each of the department-specific systems.

The TaskList class is incomplete for two main reasons. First is that there is no relationship between ProjectCoordinator and TaskList classes. The project coordinator is responsible for updating project information so they need to access the TaskList class. Second reason is that TaskList is responsible for both task handling and communication. This violates the Single Responsibility Principle. This should be split into two different classes to maintain the principle and follow software engineering principles.

Limitations:

The diagram is focused on the structural design of the Project Costing & Management System. However, it doesn't show the classes that handle the flow of data properly. An example of this is how the ProgressReport class is created by a ProjectCoordinator, and suddenly goes to the ProjectManager. The class diagram should provide another class that handles the flow of this data. This must be made clearer in the next iteration of the class diagram.

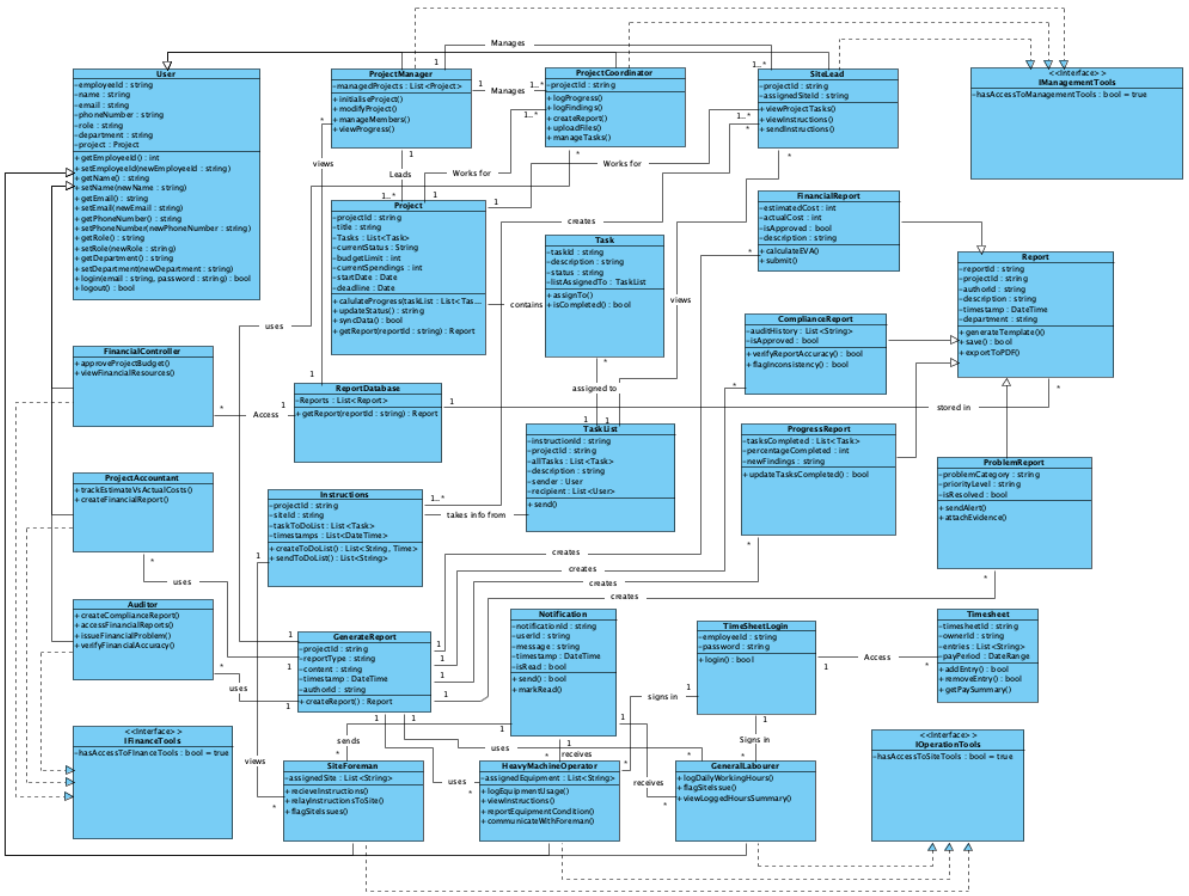
Even though security and role-based access was talked about, it is not present in the class diagram. The user class has no access or priority attribute. This is a significant limitation because the class diagram is the blueprint for the code implementation. This should be implemented into the class diagram as soon as possible, so that this doesn't become an even bigger problem later on. Another reason why this is important is because there are important documents and reports handled by the Project Manager and Financial Controller. No other user should be able to access these resources.

Design Risks:

As mentioned above, there is an absence of interface classes. The Report class is used as a base template for the 4 department-specific report classes. This violates the Open/Closed Principle. A new Interface class for Report should be added in the next interface, which could reduce coupling between the sub-classes of Report. This will allow the structure of the Reporting system to be extended without having to modify existing classes.

Most classes are loosely tied together and can fall apart easily. The second iteration should focus on tightening these relationships but in a way that doesn't make them depend on each other. The goal of the system should be to follow the four principles of abstraction, modularity, high cohesion, and low coupling.

5.3 Class Diagram - Iteration Two



This is the second iteration of the class diagram for Fulton Hogan's Project Costing & Management system. It greatly improves weaknesses and risks of the previous iteration like the security, reference attributes, interfaces, etc.

5.4 Design Weaknesses, Limitations and Risks - Iteration Two

Weaknesses:

The biggest weakness in the first iteration was that there were missing reference attributes that are necessary to support their relationships. This was dealt with by analysing each class carefully and adding in the appropriate attributes. Interfaces and inheritance is now added, as shown by the new user interfaces that each user implements based on their department. While it has improved, it will have to be further improved in order to correctly follow software engineering principles and UML modelling principles.

The TaskList and Instruction classes are unclear and have overlapping responpsibilities, violating the cohesion principle. If it is decided they do similar things, they should be merged in the next iteration. If they do different things, then they should have clear differences that follow the Single Responsibility Principle.

Limitations:

The flow of data is now shown more accurately. This interface introduced a ReportDatabase class, which makes it easier for the stakeholders to understand how reports are made and viewed by different people.

Security has been improved slightly, as shown in the TimeSheetLogin class where employeeId is now taken instead of a brand new “username” attribute. This aligns with Fulton Hogan’s existing employee records and makes the migration from their old system to this one much easier. The new user interfaces also add a layer of security by checking the user’s department. If they are in a certain department, then they get access to that department’s tools. However, this must be improved further in the next iteration of the class diagram to ensure security is there.

Design Risks:

The Reporting part of this system has been improved by adding a ReportDatabase class. This improved the connection between the Report class, and any connected classes. The Report class still doesn’t have an interface so it is still violating the Open/Closed Principles. The ReportInterface interface is yet to be added.

In the first iteration, classes were loosely tied together making the system weak and unreliable. Reference attributes have been implemented into certain classes, and new classes have been introduced to the diagram to futher strengthen the system. These improvements should be consistent through the following iterations to ensure a system that follows all software engineering principles.

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Github:
https://github.com/fatehbhular/FultonHogan-FatehBhular-23217987